CPEN 416 / EECE 571O

Lecture 16: Expectation values, mixed states, and density matrices

Tuesday 4 November 2025

Announcements

- Quiz 7 today
- Practice problem set 4 partially available
- Friday office hours cancelled due to travel

Where are we going?

Learning outcomes

- Define an observable and compute its expectation value
- Define *mixed states* and express quantum states as density matrices
- Identify mixed states on the Bloch sphere

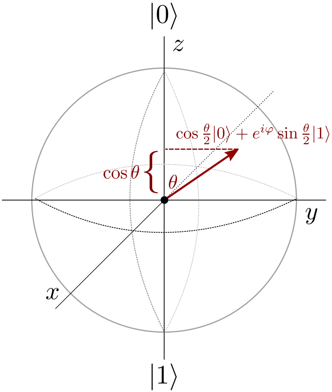
Review: projective measurements

Recall: when we measure a qubit in state $|\varphi\rangle$ with respect to basis $\{|\psi_i\rangle\}$, the probability of observing it in state $|\psi_i\rangle$ (outcome i) is

If we observe outcome i, the qubit *remains in* $|\psi_i\rangle$.

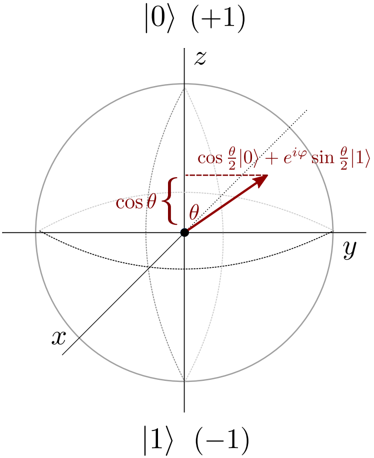
Review: projective measurements

Projective measurement corresponds to a geometric projection onto the axis represented by a basis.



Let's look at measurement in a slightly different way...

Expectation values



Assign each basis states a real-valued outcome:

If we prepare and measure *N* times, what is the **expected value** of the meas. outcome?

In the limit of infinite shots...

Expectation values

This assignment is no accident...

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Its eigensystem is

Z is an example of a more general concept: an observable.

Observables

Generally, we are interested in measuring real, physical quantities. In physics, these are called observables.

Mathematically, an observable M is a Hermitian operator (matrix)

Analytically, the **expectation value** of measuring the observable M given the state $|\psi\rangle$ is

Why Hermitian?

Observables

Example:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

X is Hermitian and its (normalized) eigensystem is

Observables

Example:

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Y is Hermitian and its (normalized) eigensystem is

Expectation values: analytical

Exercise: consider the quantum state

$$|\psi\rangle = \frac{1}{2}|0\rangle - i\frac{\sqrt{3}}{2}|1\rangle.$$

Compute the expectation value of Y:

Expectation values

Let's do this in PennyLane instead:

```
dev = qml.device('default.qubit', wires=1)

@qml.qnode(dev)
def measure_y():
    qml.RX(2*np.pi/3, wires=0)
    return qml.expval(qml.PauliY(0))
```

Multi-qubit expectation values

Example: operator $Z \otimes Z$.

Eigenvalues are computational basis states:

$$(Z \otimes Z)|00\rangle = |00\rangle$$
$$(Z \otimes Z)|01\rangle = -|01\rangle$$
$$(Z \otimes Z)|10\rangle = -|10\rangle$$
$$(Z \otimes Z)|11\rangle = |11\rangle$$

To compute an expectation value from data:

$$\langle Z \otimes Z \rangle = \frac{1 \cdot n_1 + (-1) \cdot n_{-1}}{N}$$

Multi-qubit expectation values

Example: operator $X \otimes I$.

Eigenvalues of X are the $|+\rangle$ and $|-\rangle$ states:

$$
\begin{aligned}
(X \otimes I)|+0\rangle &= |+0\rangle \\
(X \otimes I)|+1\rangle &= |+1\rangle \\
(X \otimes I)|-0\rangle &= -|-0\rangle \\
(X \otimes I)|-1\rangle &= -|-1\rangle
\end{aligned}
$$

Fun fact: All Pauli operators have an equal number of $+1$ *and* -1 *eigenvalues!*

Multi-qubit expectation values

How to compute expectation value of X from data, when we can only measure in the computational basis?

Basis rotation: apply H to first qubit

$$
\begin{aligned}
(H \otimes I)(X \otimes I)|+0\rangle &= |00\rangle \\
(H \otimes I)(X \otimes I)|+1\rangle &= |01\rangle \\
(H \otimes I)(X \otimes I)|-0\rangle &= -|10\rangle \\
(H \otimes I)(X \otimes I)|-1\rangle &= -|11\rangle
\end{aligned}
$$

When we measure and obtain $|10\rangle$ or $|11\rangle$, we know those correspond to the -1 eigenstates of $X \otimes I$.

Hands-on: multi-qubit expectation values

Multi-qubit expectation values can be created using `@` :

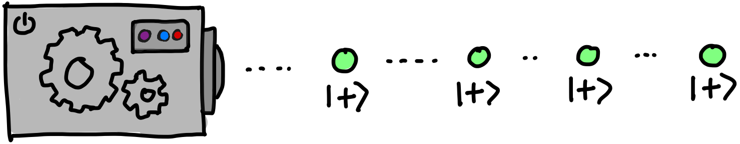
```
@qml.qnode(dev)
def circuit(x):
    qml.Hadamard(wires=0)
    qml.CRX(x, wires=[0, 1])
    return qml.expval(qml.PauliZ(0) @ qml.PauliZ(1))
```

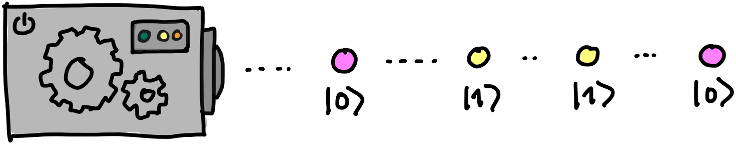
Can return *multiple* expectation values if no shared qubits.

```
@qml.qnode(dev)
def circuit(x):
    qml.Hadamard(wires=0)
    qml.CRX(x, wires=[0, 1])
    return qml.expval(qml.PauliZ(0)), qml.expval(qml.
                                                  PauliZ(1))
```

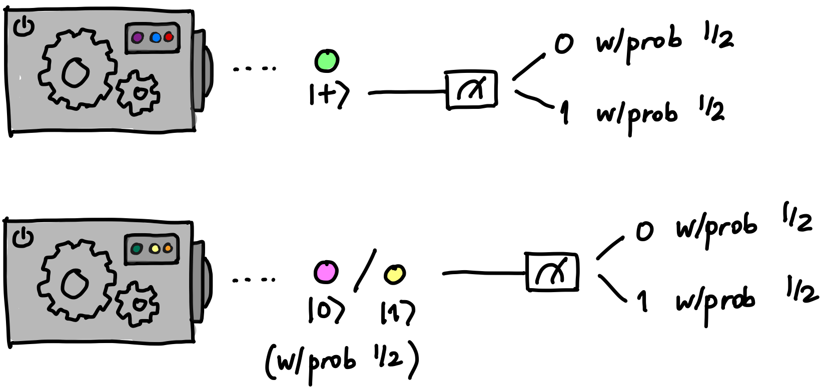
Mixed states

Are these two devices the same?

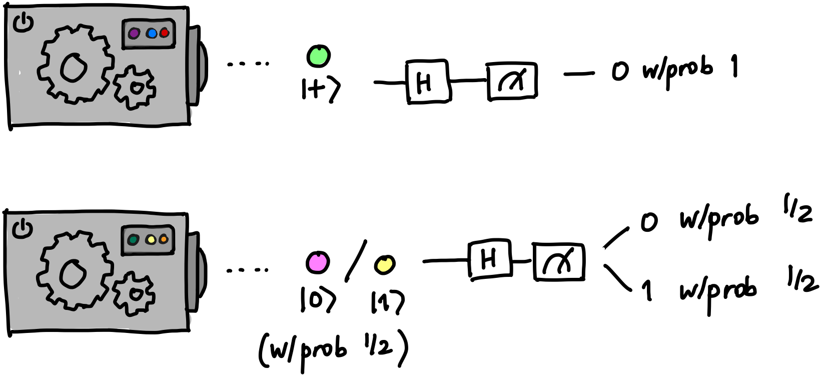




Mixed states



Mixed states



What is the second box doing?

Mixed states

The second box prepares a **mixed state**.

A **pure state** can be expressed as a single ket vector, e.g.,

A **mixed state** is a *probabilistic mixture of pure states*. It is represented by a **density matrix**.

Density matrices

The density matrix of a pure state $|\psi\rangle$ is

Exercise: what are the density matrices for $|0\rangle$ and $|1\rangle$?

Density matrices

Density matrices of mixed states are linear combinations of those of pure states:

Exercise: A system prepares $|+\rangle$ with probability $1/3$, and $|0\rangle$ with probability $2/3$. What is its state?

Density matrices

Density matrices have some nice properties.

- they are Hermitian
- they have trace 1
- they are positive semi-definite (all eigenvalues are ≥ 0)
- (for pure states only) they are projectors, i.e., $\rho^2 = \rho$

Check with our example:

Fun activity: show properties hold for general $\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i|$

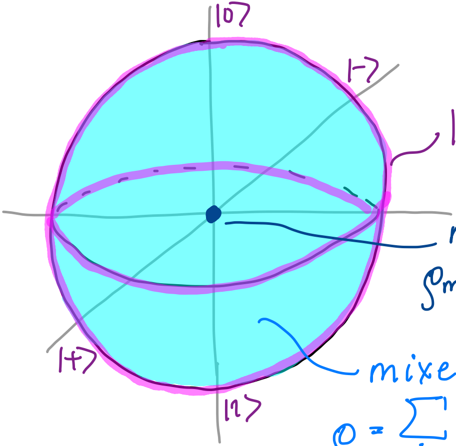
Density matrices and expectation values

Exercise: A system prepares $|+\rangle$ with probability $1/3$, and $|0\rangle$ with probability $2/3$.

$$\rho = \begin{pmatrix} \frac{5}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} \end{pmatrix}$$

What are the expectation values $\langle X \rangle$,$\langle Y \rangle$, $\langle Z \rangle$?

Density matrices and the Bloch sphere



pure states

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

maximally mixed state

$\rho_{max} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\mathbb{1}$

mixed states

$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$

Next time

Next class:

- Operations and measurements on mixed states
- Quantum channels

Action items:

1. Work on project (remember to fill out weekly surveys!)

Recommended reading:

- Codebook module NT
- Nielsen & Chuang 2.4, 2.2.5-2.2.6